

SafeSight PPE

AI-Assisted PPE Compliance Review for Construction Site Images

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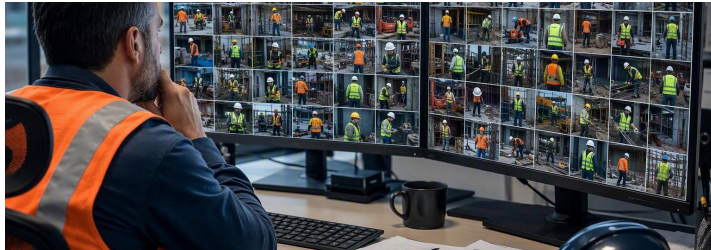
ITAI 1378 — Computer Vision and AI

Tier 2 — Advanced



The Problem

Construction safety teams need a consistent way to verify whether workers appear to be wearing required PPE such as hard hats and safety vests. Manually reviewing site photos can become time-consuming and inconsistent when there are many workers or images to review.



Who cares:

Safety managers, site supervisors, and construction companies responsible for monitoring PPE compliance.

Why it matters:

Potential PPE violations can be overlooked during manual review, especially in crowded or busy worksite images.

What could help:

A computer vision system that automatically flags possible PPE violations for human review.

Safety Manager → Site Supervisor → Construction Company

1 Worksite Image

Construction-site photo



2 YOLO Detection

Detect workers, hard hats & safety vests



3 Compliance Logic

Identify possible missing PPE

-  Count workers
-  Check hard hat status
-  Check safety vest status
-  Flag potential violations

4 Annotated Output

Image + review summary



PPE REVIEW SUMMARY

Workers detected	4
Hard hats detected	2
Safety vests detected	3
Missing hard hats	2
Missing safety vests	1


**REVIEW
REQUIRED**
3 possible PPE
violations detected

Conceptual workflow shown for illustration

Solution

SafeSight PPE analyzes construction-site images with YOLO, identifies workers and key PPE conditions, and turns those detections into an annotated image and a short compliance-review summary.

Technical Approach

Computer Vision Technique

Object Detection

Detect workers, hard hats, missing hard hats, safety vests, and missing safety vests in construction-site images.

Model -YOLO 11

Start with pretrained YOLO weights, then fine-tune the model on the Construction Site Safety dataset.

Framework- Ultralytics + PyTorch

Used for model loading, transfer learning, inference, evaluation, and annotated detection results.

Application Component Rule-Based Compliance Logic

SafeSight reads the model detections, counts possible PPE violations, and converts the raw predictions into a review summary.

Why this approach?

YOLO was selected because it can detect multiple workers and PPE items in a single image while also providing confidence scores and bounding boxes. Fine-tuning allows the pretrained model to learn the specific PPE categories needed for SafeSight, while the compliance logic turns the detections into information a safety manager can review.

Data Plan

Primary Dataset:
Construction Site Safety

Source:
Roboflow Universe

CONSTRUCTION SITE SAFETY:

- 2,801 Images
- 2,605 Train
- 114 Validation
- 82 Test

PROJECT LABELS:

-  Person
-  Hardhat
-  NO-Hardhat
-  Safety Vest
-  NO-Safety Vest

Why this dataset?

Explicit missing-PPE labels make it well suited for compliance review instead of simple PPE recognition.

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Success Metrics

1. Detection Performance

Target: $mAP50 \geq 0.70$

Measure how accurately SafeSight detects workers and PPE on unseen test images.

2. Processing Speed

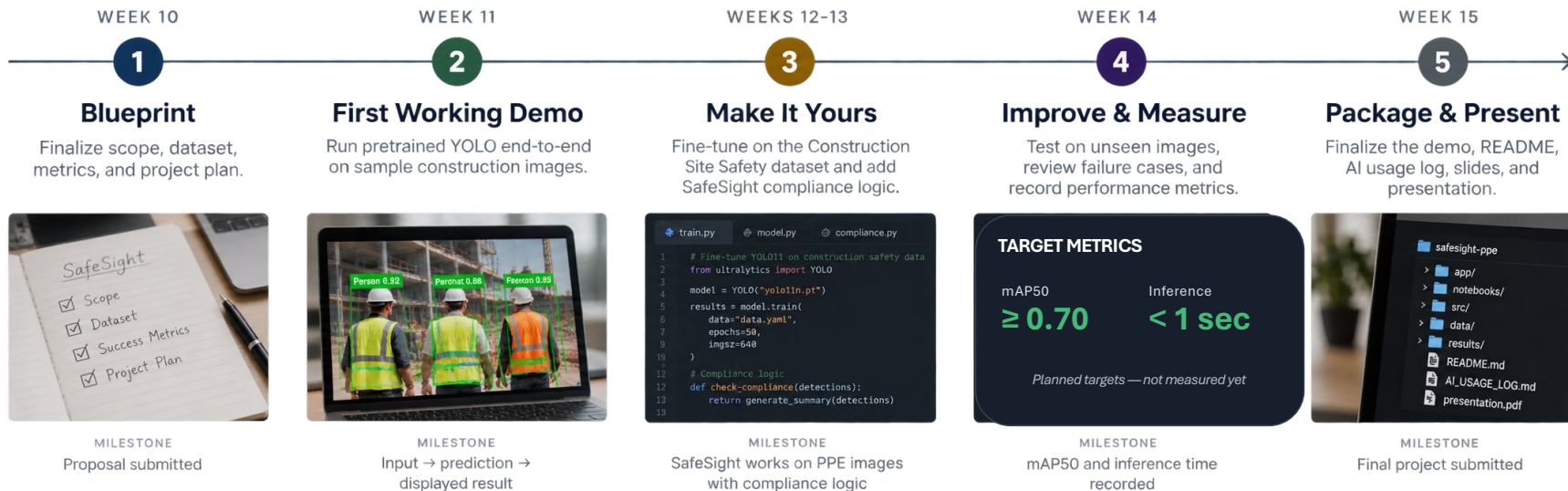
Target: < 1 second per image

Measure how quickly the system produces detections and the review summary.



Success means the system is both reasonably accurate and fast enough to support practical image review.

Milestone Plan



Concept visuals illustrate planned work and targets; final results will be measured during the Build phase.

Risks & Resources

Plan for uncertainty without overcomplicating the build.

Risk	Probability	Plan B
Model struggles to reliably detect missing PPE	Medium	Narrow the scope to hard-hat compliance first, or fall back to the baseline pretrained model while focusing on application logic and evaluation
Training/compute limitations in Colab	Medium	Reduce model size or epochs, save checkpoints to Google Drive, or switch to Kaggle/Heidi

Resources:

Compute: Google Colab

Backup Compute: Kaggle / HCC Heidi

Frameworks: Ultralytics + PyTorch

Dataset: Construction Site Safety

Estimated Cost: \$0